

Final Consulting-Style Time Series EDA Report

Professional Time Series EDA Report

Executive Summary

The dataset contains 7,423 high-frequency observations collected over approximately 147.6 seconds.

The dominant sampling interval is approximately 20 milliseconds (~50 Hz).

Key Findings:

- No missing values in data columns.
- Timestamp continuity issues due to missing and duplicate timestamps.
- Strong autocorrelation exists.
- Signal is statistically non-stationary.
- Outliers likely correspond to physiological peaks rather than corruption.

Graph 1 – Original Time Series Plot

Purpose: Visualize raw behavior of corrected red signal.

Interpretation: Oscillatory behavior is visible throughout recording with meaningful temporal structure.

Graph 2 – Cleaned Time Series Plot

Purpose: Visualize signal after preprocessing.

Interpretation: Shape remains unchanged because no significant missing values existed.

Graph 3 – Time Series Decomposition

Trend: Mild baseline drift.

Seasonality: Strong repeating physiological patterns.

Residual: Remaining noise and sensor artifacts.

Graph 4 – Monthly Boxplot

Dataset duration is too short for meaningful monthly analysis.

Graph 5 – Weekly Boxplot

Dataset duration is too short for meaningful weekly analysis.

Graph 6 – Outlier Visualization

Outliers are largely associated with physiological peaks.

Graph 7 – ACF Plot

Strong positive autocorrelation at low lags, indicating temporal dependence.

Graph 8 – PACF Plot

Strong lag-1 influence followed by gradual decline.

Stationarity Analysis

ADF Statistic: -2.5364

p-value: 0.1069

Conclusion: Non-stationary series.

Descriptive Statistics

Count: 7423

Mean: 7.390

Std Dev: 88.545
Minimum: -232.530
Median: -13.113
Maximum: 441.644

Final Findings

The dataset represents a clean, high-frequency physiological time series with strong temporal structure and meaningful periodic behavior. The primary limitation is the short recording duration (~148 seconds), preventing meaningful monthly and weekly seasonality analysis.